

Side-Channel Assisted Analysis of the Code Equivalence Problem: Attacking the LESS Signature Scheme via Noisy Hints

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Abstract—Recent advancements in quantum computing have necessitated the transition toward post-quantum cryptography (PQC), with code-based signatures emerging as a robust alternative. Among these, the Linear Equivalence Signature Scheme (LESS) stands out due to its security reliance on the hardness of the Linear Equivalence Problem (LEP). However, physical implementations of such schemes often introduce unintended vulnerabilities through side-channel leakages. In this work, we investigate the susceptibility of LESS to side-channel assisted cryptanalysis. Specifically, we exploit a bitwise Bayesian leakage model to extract noisy hints regarding the secret permutation and scaling factors. To process and exploit this side-channel information, we propose an algorithmic framework that directly integrates these probabilistic hints into the decoding and resolution process of the LEP. Our approach leverages the structural properties of the code-equivalence problem to drastically prune the search space. Experimental results demonstrate that leveraging physical leakage through our side-channel-aware algorithm significantly reduces the security margin of LESS, providing a new perspective on the practical resistance of code-equivalence-based primitives.

Index Terms—Post-Quantum Cryptography, LESS, Linear Equivalence Problem, Side-Channel Attacks, Bayesian Leakage, Cryptanalysis, Guided Decoding

I. INTRODUCTION

The emerging era of quantum computing represents one of the most significant challenges to global digital security. Theoretically, large-scale quantum computers could leverage Shor’s algorithm to solve integer factorization and discrete logarithm problems in polynomial time, effectively breaking the cryptographic foundations of RSA, Diffie-Hellman, and Elliptic Curve Cryptography (ECC). In light of this, the National Institute of Standards and Technology (NIST) initiated a multi-year global effort to evaluate and standardize Post-Quantum Cryptographic (PQC) primitives. These new standards must demonstrate resilience not only against classical cryptanalysis but also against adversaries equipped with quantum capabilities.

Within the landscape of PQC, code-based cryptography has stood the test of time, dating back to the McEliece cryptosystem in 1978. While the majority of these schemes rely on the NP-hardness of the Syndrome Decoding Problem (SDP), an expanding branch of research focuses on the structural properties of linear codes. The Linear Equivalence Signature Scheme (LESS), currently

under evaluation in the NIST PQC process, departs from the traditional SDP paradigm. Instead, its security is anchored in the Linear Equivalence Problem (LEP) and the Permutation Equivalence Problem (PEP). These problems task an adversary with discovering a secret transformation—consisting of a permutation matrix and a non-singular diagonal scaling matrix—that maps one public linear code to another.

Despite the proven theoretical hardness of the LEP in the standard model, the transition from mathematical theory to physical implementation often introduces unforeseen vulnerabilities. Modern cryptographic hardware is susceptible to side-channel attacks (SCA), where physical observables such as power consumption, electromagnetic radiation, or thermal variations leak partial information about the internal state of the device. Traditionally, cryptanalytic solvers for the LEP/PEP, such as Leon’s algorithm or Beullens’ recent improvements, operate under a “zero-knowledge” assumption, where the adversary has no a priori information regarding the secret transformation. However, this assumption is often too optimistic for real-world deployments.

This paper addresses the critical need to quantify the security of LESS in the presence of physical leakages. We bridge the gap between abstract complexity and practical security by introducing a framework that accounts for partial, noisy information. Central to our study is the Bitwise Bayesian Leakage Model (BBLM). This model, frequently applied in the context of cold boot attacks and side-channel analysis, provides a probabilistic description of the secret keys. By representing the leaked information as “noisy hints” or posterior probabilities, we can model an adversary who possesses an imperfect but useful view of the secret permutation and scaling factors.

The main contribution of this work is the design and implementation of an algorithmic recovery framework that integrates these Bayesian hints directly into the resolution process of the LEP. Unlike classical solvers that must explore the entire combinatorial space of permutations, our side-channel-aware algorithm uses the posterior distributions to prune the search tree and prioritize the most likely candidates. Our results demonstrate that even low-fidelity leakage can significantly erode the security margins of LESS, suggesting that the “hardness” of the code equivalence problem must be re-evaluated when the hardware—often the weakest link in the cryptographic

chain—is taken into account.

II. BACKGROUND

A. Code-Based Cryptography

A linear code $\mathcal{C}$ of length n and dimension k over a finite field $\mathbb{F}_q$ is a k -dimensional subspace of $\mathbb{F}_q^n$. The elements of $\mathcal{C}$ are called codewords. A code is typically defined through a generator matrix:

Definition II.1 (Generator Matrix). A matrix $G \in \mathbb{F}_q^{k \times n}$ is a generator matrix for a linear code $\mathcal{C}$ if its rows form a basis for $\mathcal{C}$. The code is defined as the set of all possible linear combinations of these rows:

$$\mathcal{C} = \{\mathbf{u}G \mid \mathbf{u} \in \mathbb{F}_q^k\}. \quad (1)$$

In schemes such as LESS, public and secret keys are represented by these matrices. The security of the primitive is rooted in the difficulty of recovering the secret transformation between two equivalent generator matrices.

B. Code Equivalence Problems

The equivalence between codes is defined by the existence of a transformation that preserves the code's structural properties, such as its Hamming weight. Algebraically, this involves both a column transformation (permutation or scaling) and a potential change of basis for the row space.

Definition II.2 (Permutation Code Equivalence Problem - PEP). Given two generator matrices $G_1, G_2 \in \mathbb{F}_q^{k \times n}$ of two permutation equivalent codes, find a permutation matrix $P \in \mathbb{F}_q^{n \times n}$ and an invertible matrix $S \in GL_k(\mathbb{F}_q)$ such that:

$$G_2 = SG_1P. \quad (2)$$

Definition II.3 (Linear Code Equivalence Problem - LEP). Given two generator matrices $G_1, G_2 \in \mathbb{F}_q^{k \times n}$ of two linearly equivalent codes, find a monomial matrix $M \in \mathbb{F}_q^{n \times n}$ and an invertible matrix $S \in GL_k(\mathbb{F}_q)$ such that:

$$G_2 = SG_1M. \quad (3)$$

A monomial matrix M is defined as a matrix with exactly one non-zero entry in each row and column, representing the composition of a permutation and a scaling of the code's coordinates.

C. Threat Model: Bitwise Bayesian Leakage

We formalize the threat model used throughout this work, termed the *Bitwise Bayesian Leakage Model* (BBLM). The BBLM provides a general probabilistic abstraction for modeling information leakage at the level of individual secret bits. This model captures leakage through per-bit likelihood functions and the corresponding posterior distributions, enabling principled reasoning about key recovery across scenarios such as memory decay or side-channel inference.

Formally, let $\mathbf{x}^* = (x_1^*, \dots, x_N^*) \in \{0, 1\}^N$ denote the unknown secret bit string representing the transformation. The adversary obtains leakage observations $\mathcal{L} = (L_1, \dots, L_N)$. For each bit position i , the adversary computes the posterior distribution conditioned on L_i . By Bayes' rule:

$$\Pr(x_i^* = b \mid L_i) = \frac{\Pr(L_i \mid x_i^* = b) \Pr(x_i^* = b)}{\Pr(L_i)}, \quad (4)$$

where $\Pr(x_i^* = b)$ is the prior probability. To ensure tractable inference, we adopt the standard assumptions of *bitwise independent prior* and *conditional independence of leakage*, allowing the posterior distribution to factorize:

$$\Pr(\mathbf{x}^* \mid \mathcal{L}) = \prod_{i=1}^N \Pr(x_i^* \mid L_i). \quad (5)$$

1) *Leakage Instantiations*: The BBLM encompasses several important leakage models:

- **Erasur Setting**: The adversary learns a subset of bits exactly, while the rest are erased ($L_i = \perp$), yielding $\Pr(x_i^* \mid \perp) = 1/2$ under uniform priors.
- **Error Setting**: Corresponds to a binary symmetric channel with crossover probability δ , where $\Pr(x_i^* \neq L_i \mid L_i) = \delta$.
- **Cold Boot Attacks (CBA)**: Models asymmetric bit-flip probabilities (α for $0 \rightarrow 1$ and β for $1 \rightarrow 0$) in DRAM. Typically, $\alpha \ll \beta$.
- **Side-Channel Classifiers**: Classifiers implementing functions $g_i(L_i) \approx \Pr(x_i^* = 1 \mid L_i)$ from physical measurements.

2) *Posterior Scoring*: Given a candidate bitstring $x' \in \{0, 1\}^N$, we utilize the additive scoring rule:

$$\text{Score}(x') = \sum_{i=1}^N \log \Pr(x_i^* = x'_i \mid L_i). \quad (6)$$

Maximizing this score is equivalent to maximum a posteriori (MAP) estimation, which guides the search toward the most probable secret matrix.

D. Setting for the CBA in Code Equivalence Problem

To bridge physical leakage with algebraic cryptanalysis, we specialize the BBLM to the structure of the *Code Equivalence Problem*. We consider an adversary seeking to recover the secret monomial matrix $M \in \mathbb{F}_q^{n \times n}$ stored in memory or processed during the execution of a LESS signature.

Let $\mathbf{x}^*$ be the binary representation of the entries of M . We define a *noisy candidate matrix* M' whose components are derived from the MAP estimation of the leaked bits. The recovery task is reformulated as finding the exact transformation M that satisfies:

$$G_2 = S \cdot (G_1M), \quad (7)$$

where G_1, G_2 are the public generator matrices and S is the unknown change-of-basis matrix. This approach transforms the purely combinatorial search over the $n!(q-1)^n$ space into a *guided search* over candidates prioritized by their Bayesian scores.

III. STATE OF THE ART

The security landscape of the Code Equivalence Problem (CEP) has transitioned from theoretical interest to a critical area of study within Post-Quantum Cryptography (PQC). This section provides a formal analysis of the algorithmic evolution from classical combinatorial solvers to modern collision-based attacks, as well as the current status of the LESS signature scheme and its variants.

A. Evolution of Combinatorial Solvers: Leon and SSA

The foundational approach to solving the Permutation Equivalence Problem (PEP) was introduced by Leon [?]. This algorithm is based on the principle that any equivalence between two linear codes must preserve the weight distribution of their respective subspaces. Leon's method involves the construction of a *representative set* of low-weight codewords in both G_1 and G_2 . By analyzing the supports of these codewords, the algorithm derives a system of linear constraints intended to recover the secret permutation π .

Subsequently, Sendrier proposed the *Support Splitting Algorithm* (SSA) [?]. SSA introduced the concept of a *coordinate signature*, an algebraic invariant that remains stable under permutation. While SSA demonstrates polynomial-time efficiency for random codes over $\mathbb{F}_2$, its discriminative power is significantly reduced over larger fields $\mathbb{F}_q$ ($q > 2$). In such cases, the complexity of computing higher-order invariants for the Linear Equivalence Problem (LEP) renders the algorithm less effective against modern primitives.

B. Collision-based Cryptanalysis: The Beullens Algorithm

The current benchmark for solving equivalence problems is the algorithm developed by Beullens [?]. This approach deviates from previous techniques by utilizing a *meet-in-the-middle* strategy focused on identifying collisions between normalized codewords. The algorithm employs a modified *Information Set Decoding* (ISD) procedure to sample codewords from $\mathcal{C}_1$ and $\mathcal{C}_2$ that satisfy specific structural constraints.

The methodology is specialized for the two primary variants of the problem:

- **Framework for PEP:** The algorithm generates two sets of candidate codewords, $\mathcal{L}_1 \subset \mathcal{C}_1$ and $\mathcal{L}_2 \subset \mathcal{C}_2$, characterized by a fixed weight or support. A collision search is performed to identify a pair (c, c') such that $c' = \pi(c)$, which effectively reveals the underlying permutation.
- **Framework for LEP:** To account for the unknown scaling factors $\sigma_i \in \mathbb{F}_q \setminus \{0\}$, a *normalization procedure* is implemented. Each sampled codeword is scaled by the inverse of its first non-zero entry, ensuring that linearly equivalent codewords yield identical normalized representatives. This transformation maps the LEP instance into a permutation-search space.

Beullens' cryptanalysis demonstrated that the initial security parameters for LESS were overestimated. For

instance, the LESS-III instance, intended for category 1 security, was solved in approximately 2 seconds. The computational cost of the attack is defined by:

$$\text{Cost} \approx \frac{\sqrt{|\text{Aut}(\mathcal{C})| \cdot q^k}}{N_w}, \quad (8)$$

where N_w denotes the density of available codewords of weight w and $|\text{Aut}(\mathcal{C})|$ represents the order of the automorphism group of the code.

C. The LESS Signature Scheme and its Variants

The Linear Equivalence Signature Scheme (LESS) has been refined to mitigate the efficiency of collision-based attacks. The scheme utilizes a Sigma-protocol for LEP, transformed into a non-interactive signature through the Fiat-Shamir heuristic.

- **Standard LESS:** The original submission to NIST [?] established parameter sets over fields $\mathbb{F}_q$ (with $q \in \{127, 251\}$), typically with length $n \approx 128$ and dimension $k \approx 64$.
- **LESS-FM (Fine-tuning):** This variant, introduced by Barengi et al. [?], incorporates *canonical forms* for response compression. By representing codes in a fixed row-echelon form, the scheme reduces the public key size and complicates the normalization required for collision attacks.

Current security evaluations assume a *zero-knowledge* environment, where the adversary possesses no prior information regarding the secret transformation matrix M .

D. Open Challenges and Physical Security

A significant research gap exists concerning the security of these primitives under physical leakage. Conventional solvers do not incorporate auxiliary information potentially available during the execution of the signature.

PENDING: Investigar literatura sobre Side-Channel Attacks y Machine Learning aplicados a firmas basadas en códigos.

IV. CONCLUSIONS AND FUTURE WORK REFERENCES